Hong-Son Nguyen

Nguyễn Hồng Sơn

Robotics / AI Researcher — autonomous navigation, locomotion, perception, reinforcement learning

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Profile

Robotics researcher who ships complete autonomy stacks, not just models. Three years building navigation systems that run on real hardware — perception, prediction, planning, learning and control — for mobile robots operating around people. Seven publications including IEEE ICRA 2026 (A*-ranked in Robotics and Automation).

Education

2024 – present

M.S. in Robotics and Control Systems

National Cheng Kung University, Taiwan (#191 QS Ranking) — GPA 3.52/4.0

2019 – 2023

B.S. in Mechatronic Engineering

Hanoi University of Industry, Vietnam — GPA 3.05/4.0

Technical skills

Languages	Python, C++, MATLAB
Machine Learning	PyTorch, deep reinforcement learning, Attention mechanism, selective state space models (Mamba), Sim2Sim and Sim2Real transfer
Libraries	ROS 2, NVIDIA Isaac Sim / IsaacLab, Gazebo, MuJoCo, CUDA
Hardware	Agilex Scout Mini, Unitree Go2, Jetson AGX Orin, edge deployment

Experience

Feb 2024 – present

Research Fellow — Robotics

Networked Robotic Systems Laboratory, National Cheng Kung University, Taiwan

- Own an end-to-end human-aware navigation stack running on an Agilex Scout Mini: LiDAR and stereo perception, pedestrian tracking, trajectory prediction, planning and control, deployed on Jetson AGX Orin.
- Designed Social Graph Mamba, a trajectory forecasting model that replaces quadratic attention with selective state space models over dynamic interaction graphs, giving linear scaling in the number of tracked people while matching benchmark accuracy on ETH/UCY and SDD.
- Built large-scale RL training environments in Isaac Sim / IsaacLab with configurable crowds, obstacles and curricula, and trained navigation policies with PPO via RSL-RL.
- Designed and ran human-subject experiments with real pedestrians, turning comfort surveys and their statistical analysis into an empirical model used by the planner. Four papers in two years, one at IEEE ICRA 2026.

Funded projects: *Advanced Resilient and Intelligent Control for Heterogeneous Cyber-Physical Mobile Robot Systems*, MOST, Taiwan (Mar – Jul 2024); *Designing and Implementing Generative AI Service Robots: Exploring Applications in Companionship and Home Services*, NSTC, Taiwan (Aug 2024 – Aug 2025); *Trust-Integrated Intelligent Control for Socially Aware Robot Navigation in Dynamic and Crowded Environments*, NSTC, Taiwan (Aug 2025 – present).

2022 – Jan 2024

Research Assistant, Team Leader

Hanoi Intelligent Robotics Laboratory, Hanoi University of Industry, Vietnam

- Led a student research team on reinforcement learning for autonomous mobile robots, coordinating the work that produced the group's first published navigation papers.
- Implemented and benchmarked Q-Learning, SARSA and a deep RL / Pure-Pursuit hybrid controller in Gazebo, with validation on physical robots.

Funded project: *Development of the Localization System for Mobile Robots Working in the Outdoor Environment*, Hanoi University of Industry (HAUI), Vietnam (Dec 2022 – Dec 2023).

Selected projects

2026

Social Graph Mamba

Attention-free multi-agent trajectory prediction

Selective state space models over dynamically built interaction graphs, with a community-aware module using differentiable MinCut to recover pedestrian groups. Linear sequence complexity instead of the quadratic cost of attention, evaluated on ETH/UCY and SDD and deployed on a physical robot.

2025

RL Social Navigation

Large-scale training environment in NVIDIA IsaacLab

Configurable crowds, obstacles, curricula and goal distributions for learning socially aware navigation policies, trained with PPO through RSL-RL and transferred to the Scout Mini platform.

2023

Autonomous Hexapod Robot

Undergraduate thesis — graded 94, selected as best thesis

Deep reinforcement learning for locomotion and outdoor autonomy on an 18-DOF hexapod, from simulation training through to hardware deployment.

Additional personal projects are documented on GitHub: sontypo.

Publications and awards

Publications Seven papers on robot navigation and pedestrian prediction, including IEEE ICRA 2026 and two Springer volumes — Google Scholar.

Awards Distinguished International Student Scholarship, National Cheng Kung University (2024–2026); first prize, Students Scientific Research Competition, Hanoi University of Industry (2022); departmental scholarship (2023); ASME student member since 2022.

Licenses & Certifications

2025 **Best Session Presentation Award** — International Automatic Control Conference (CACS)

2022 **Fundamentals of Reinforcement Learning** — Coursera

2022 **Supervised Machine Learning: Regression and Classification** — Coursera

2022 **Mathematics for Machine Learning: Linear Algebra** — Coursera

2021 **Python Developer for AI Course** — VTC Academy

Languages

English IELTS 6.0